7 (a) Write 0.005 49 in standard form.

5.49×10^{-3}

1 mark
Move decimal3 places

(1)

(b) Find the value of $(8 \times 10^3)^2$
Give your answer in standard form.

$8^2 \times (10^3)^2 = 64 \times 10^6$

1 mark
Square both factors

$= 6.4 \times 10^7$

1 mark
6.4×10^7

(2)

(c) Find the value of $(7.6 \times 10^5) + (8.7 \times 10^4)$
Give your answer in standard form.

$7.6 \times 10^5 + 0.87 \times 10^5$

1 mark
Use same power10

$= (7.60 + 0.87) \times 10^5$

$= 8.47 \times 10^5$

1 mark
8.47×10^5

(2)

(Total for Question 7 is 5 marks)